

AE-420: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-382-3) 365 nm and white, 50 mM KOH in D₂O

Date: 2025-01-20

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P 1H-DOSY

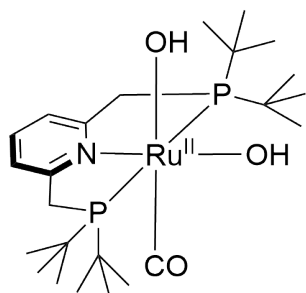
Category: Two-photon

Status: Done

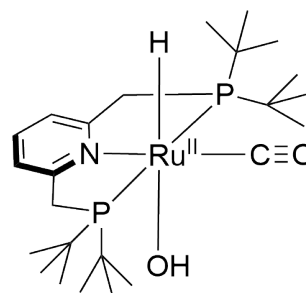
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



365 nm (0.3 W)
white (0.2 W)
H₂O



Chemical Formula: C₂₄H₄₅NO₃P₂Ru

Molecular Weight: 558,64552

[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru

Molecular Weight: 542,64652

[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	Two-photon - AE-377: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH in D ₂ O
Related experiment	Two-photon - AE-386: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D ₂ O I

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	Two-photon - AE-382: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h	/	/	/	approx. 2.5	558.65	/
KOH (0.1 M)	Two-photon - AE-375: Stock solutions of Na ₂ PO ₄ and Na ₂ CO ₃ in H ₂ O and KOH in D ₂ O (AE-375-3)	/	/	/	/	/	0.9

D2O	AE-352: Degassing of D2O	/	/	/	/	/	0.9
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Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25
Irradiation duration [h]	5

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
20.01		A 10 mL flask was prepared	
	9:15	To the flask [Ru] was added under argon counterflow	
		The flask was set under argon	
	9:20	To the flask water was added using a 1 mL syringe	after addition of H2O.jpg brownish suspension
	9:20	The obtained mixture was stirred at 600 rpm.	
	9:35	To the flask KOH (0.1 M) was added using a 1 mL syringe	after addition of KOH.jpg no significant change
	9:37	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-420-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	AE-420-1.jpg clear slightly yellow solution
21.01	9:30	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup.jpg

		The NMR-tube was placed in the setup accroding to [Protocol] Irradiation setup	before irrad.jpg no significant change
	9:39 - 14:35	The J.Young NMR tube was irradiated for 5:00 h	during irrad 1.jpg during irrad 2.jpg less light in photo
		An NMR was measured (AE-420-2)	

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
20.01	13:56	AE-420-1	1H, 1Hzgespg	IOMC 600 MHz	D2O	AE-420-1.zip	AE-420-1_10.nmrium	/	1H as expected. In 31P much more signals as expected. Looks basically identical to spectra of AE-420-2. No idea why. Maybe wrongly uploaded, since 1H looks good???

21.01	15:01	AE-420-2	1H, 1Hgzespq, 31P{1H}quant, 1H-DOSY, 31P{1H}-1H-HMBC	IOMC 600 MHz	D2O	AE-420-2.zip	AE-420-1_105.nmrium	Two-photon - AE-386: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D2O I (AE-386-2, -3) Two-photon - AE-377: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH in D2O (AE-377-2) Two-photon - AE-358: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH (AE-358-2) Two-photon - AE-348: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, water, longer irradi (AE-348-1-2)	31P: No significant differences to AE-386-3, besides two peaks observed at 90.5 (cis-hydrido-hydroxo signal) --> maybe dimerization? In comparison to AE-377-2 and AE-358-2 some more side products are observed, but with low intensities. They were not observed prior, since AE-420 has a higher signal-to-noise ratio. As a reminder: in 50 mM KOH more sideproducts form generally compared to irradiations in pure H2O. DOSY: hydrodynamic radius was not calculated, since signal is basically identical with AE-386-3 at -9.52 HMBC: wrong trace for 1H is displayed, this makes evaluation more complex: From highest to lowest shift: cis-hydrido-hydroxo: only species with weak signals outside the methyl area: very weak in aromatic area at 7.62 ppm and weak doublet at -6.00 ppm (hydride signal). Two signals in methyl area: 1.34 and 1.19 ppm. Likely corresponds to triplets observed in 1H at 1.48 and 1.34 ppm. Info: fit to cis-hydrido-hydroxo (unsymmetric and hydride signal) trans-hydrido-hydroxo: no hydride signal observed, could be due to lower concentration. In methyl 2 signals as expected at 1.24 and 1.11 ppm. Signal at 1.24 likely corresponds to triplet at 1.39 in 1H. Other triplet not seen. unreacted substrate: not as expected, only one signal at 1.11 ppm, only 3 %, maybe another minor impurity main side product: one signal observed at 1.14 ppm. in 1H signal cannot be attributed, since overlapped with many other signals. Since only one signal -> symmetric, Since observed to react back to starting material: most likely trans-dihydroxo other side products: for each only one signal is observed, but no identification possible
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Results

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product) [%]	Yield (soluable side products) [%]	Yield (other side products) [%]	Data
AE-420-2	26	6	2	25	21	20	AE-420_yield_determination.xlsx

Annotation: No 31P measured before irradiation, the other side products were determined from ¹H against the whole aromatic area.

Yields fit to expected. Side products as well.

DOSY confirms: all observable products are monomeric.

HMBC: For cis-hydrido a aromatic and a hydride signal were observed. Only both hydrides show two signals in methyl area. "cis-dihydroxo" only shows one signal (not as expected: two) --> maybe the small signal is another side product.

Main side product has only one signal, since monomeric and reacts back to starting material --> likely trans-dihydroxo

Linked experiments

- [AEI-022: Radiation of \[Ru\(CO\)\(OH\)₂\(PNN\)\], NO₂-route, 365 nm and white LED](#)
- [AEI-026: Capillary production](#)
- [AEI-064: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag₂O](#)
- [AEI-067: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], crude, 365 nm and white](#)
- [AEI-071: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], purer, 365 nm and white](#)
- [AE-105: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag₂O](#)
- [AE-110: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], purified, 365 nm and white](#)
- [AE-119: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], purified, 52 mM KOH, 365 nm and white](#)
- [AE-140: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], purified, 365 nm and white, longer duration I](#)
- [AE-145: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag₂O](#)
- [AE-158: Radiation of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\], purified, 365 nm and white](#)

- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-331: NMR of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ (AE-310-2) at lower and higher Temperature

- AE-352: Degassing of D_2O

Prep work - AE-361: 5 mM Stocksolution of KOH in water

Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation

Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-362: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM KOH

Two-photon - AE-363: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1 mM KOH

Two-photon - AE-371: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 0.1 mM KOH

Two-photon - AE-375: Stock solutions of Na_2PO_4 and Na_2CO_3 in H_2O and KOH in D_2O

Two-photon - AE-376: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM Na_2CO_3

Two-photon - AE-377: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH in D_2O

Two-photon - AE-382: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Two-photon - AE-386: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-382) 365 nm and white, 50 mM KOH in D_2O

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-420-1_105.nmrium

sha256: 7d6f3dc526309eb10ec661e83948ac3eebdd3fdf7c6f130f0771b645a7090898

AE-420_yield_determination.xlsx

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AE-420-1_10.nmrium

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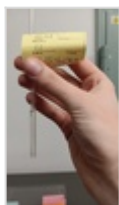
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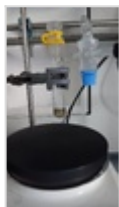
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after-addition-of-H2O.jpg

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after-addition-of-KOH.jpg

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power-setup.jpg

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before-irrad.jpg

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during-irrad-2.jpg

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during-irrad-1.jpg

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Unique eLabID: 20250120-23c8ba0ef2d6e573f6ccb1c8a0d3a92079b65d91
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1648>